# AYUSHI RAJ

**AI Training & Code Evaluation Specialist | AST Compilers & Code Quality Benchmarks**

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## PROFESSIONAL SUMMARY

Technical Developer and Code Evaluation Specialist, experienced in Abstract Syntax Tree (AST) parsing, automated test generation, and code quality evaluation. Skilled in assessing code logic, identifying runtime edge cases, and authoring automated test suites in xUnit, PyTest, and Vitest.

## TECHNICAL SKILLS

- **Autonomous AI Agents & Intelligent Systems:** Autonomous Browser Agents, Google GenAI SDK, Multi-Model LLM Routing (Groq/Gemini), Dynamic Job-Tailoring Algorithms, ATS Scoring Systems
- **System Design & Full-Stack Architecture:** Next.js 16 (App Router), React 19, Server Components (RSC), Server Actions, RESTful APIs, State Machines (Zustand), Turso LibSQL, Redis 7 Caching
- **Browser Automation & Extensions:** Chrome Extensions (Manifest V3), Shadow DOM Injection, Content Script Orchestration, Playwright, Puppeteer, DOM Parsing, Web Workers
- **Languages & Core Runtimes:** TypeScript, JavaScript, Python, C# (.NET Core), Java (Spring Boot), Node.js, Fastify, SQL
- **UI Engineering & Design Systems:** Tailwind CSS, Vanilla CSS (Glassmorphism & Design Tokens), Monaco Editor, Lucide Icons, Print-Perfect A4 Layouts, Responsive Design
- **Security, Testing & Cloud:** Web Crypto API (AES-GCM, PBKDF2), HMAC-SHA256, JWT, Vitest, Jest, PyTest, Oracle Cloud (OCI), Docker, Git/GitHub

## TECHNICAL PROJECTS

**Astra Vision - AI Sandbox & Code Parsing Platform** (FastAPI, Python, Monaco, Tree-Sitter) Live Demo | GitHub

- **Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- **Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- **Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

**AstreWork — Autonomous Career Platform & AstrePilot Extension** (Next.js 16, Chrome MV3, Gemini API) Live Demo | GitHub

- **Co-Developer & Lead Frontend Architect:** Engineered client-side ATS match engine and Gemini-powered scoring studio on Next.js 16, achieving sub-second evaluation across live postings.
- **AstrePilot Chrome Extension (Manifest V3):** Built a cross-browser extension with an isolated glassmorphic Shadow-DOM HUD and HMAC-SHA256 token authorization, executing 1-click ATS application autofill across Greenhouse, Lever, and Ashby in ~15 seconds per application.
- **High-Density Reactive UI Architecture:** Engineered a responsive, accessible workspace in Next.js 16 App Router and React 19 with Zustand state management, real-time ATS audit checklists, and print-perfect A4 CSS layout rendering with zero layout shift.

**Alyra Lock - Secure Zero-Knowledge Password Vault** (React, TypeScript, Express, MongoDB) Live Demo | GitHub

- **Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- **Cryptographic Architecture:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- **Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

## TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

**Commonwealth Bank (CommBank) - Software Development Virtual Simulation** June 2026

- **Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- **Action Taken:** Extended C# Web API controllers with $set atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- **Key Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

**Y Combinator (YC Startup - Shiptivity) - Software Development Virtual Simulation** June 2026

- **Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- **Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- **Key Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

**Walmart USA - Advanced Software Development Virtual Simulation** May 2026

- **Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- **Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (<<, >>>) and built 3NF database schemas with design pattern abstractions.
- **Key Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using csv and sqlite3 to clean and consolidate disparate spreadsheets.

## EDUCATION

**Bachelor of Computer Applications (BCA) - Amity University Noida** Expected July 2028

**Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)